Lecutre 2: ML Basics from Representation Learning view

Today:

00:00) Supervised learning formalized

00:10) Linear regression & normal equations

00:20) Gradient descent

00:45) ---- Break ----

01:00) Tensor operations

01:45) ---- Break ----

02:00) Feature transform & nonlinearity

02:20) Error decomposition & fitting

02:45) ---- Break ----

03:00) Tutorial (**Oren**): Linear Classification

Supervised Learning

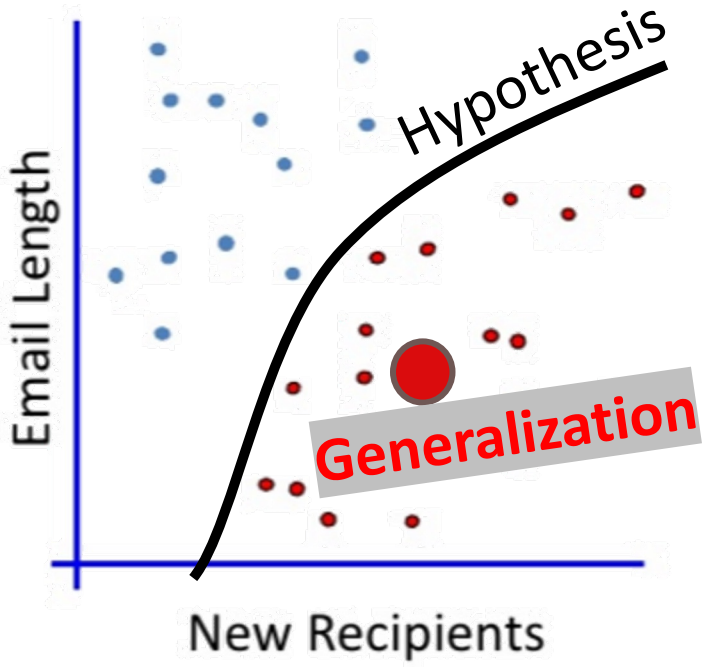
Features

Labels

Instances

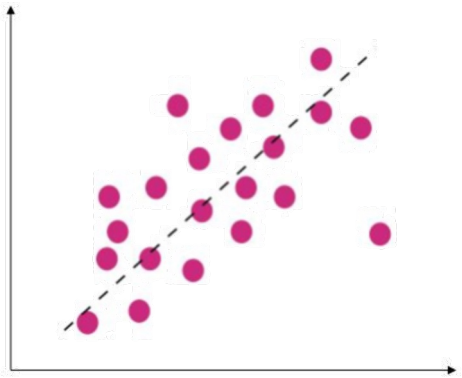
Number of new Recipients	Email Length (K)	Country (IP)	Customer Type	Email Type
0	2	Germany	Gold	Ham
1	4	Germany	Silver	Ham
5	2	Nigeria	Bronze	Spam
2	4	Russia	Bronze	Spam
3	4	Germany	Bronze	Ham
0	1	USA	Silver	Ham
4	2	USA	Silver	Spam

Numeric Nominal Ordinal

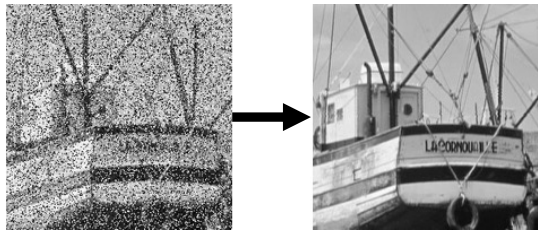


Supervised Learning

Regression

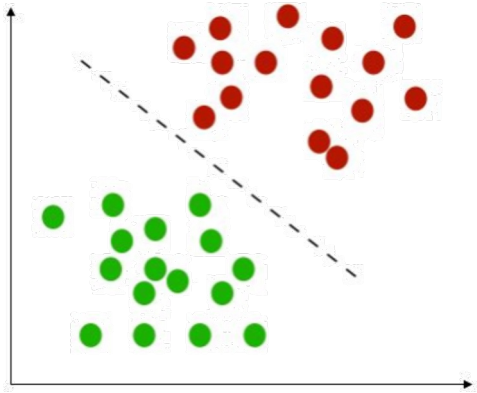


E.g. Image denoising



E.g. Object localization

Classification



E.g. Image classification

Supervised Learning

Learning Algorithm — Training set — Hypothesis

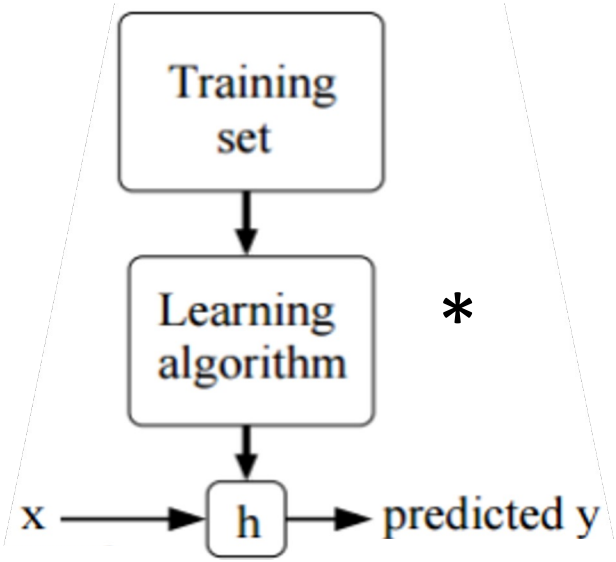
$$A(S) = h$$

Hypothesis class
$\mathcal{H} = \{h_1, h_2 \dots\}$

$h(x) \approx y$

Loss
$\mathcal{L}$

Optimization method



*

Inputs
$\boldsymbol{X}$

Labels
$\boldsymbol{Y}$

$$\begin{bmatrix} \text{———} & \mathbf{x_1}^T & \text{———} \\ \text{———} & \mathbf{x_2}^T & \text{———} \\ & \vdots & \\ \text{———} & \mathbf{x_M}^T & \text{———} \end{bmatrix}, \quad \begin{bmatrix} \text{—} & \mathbf{y_1}^T & \text{—} \\ \text{—} & \mathbf{y_2}^T & \text{—} \\ & \vdots & \\ \text{—} & \mathbf{y_M}^T & \text{—} \end{bmatrix}$$

* Diagram by Andrew NG

Normal Equations

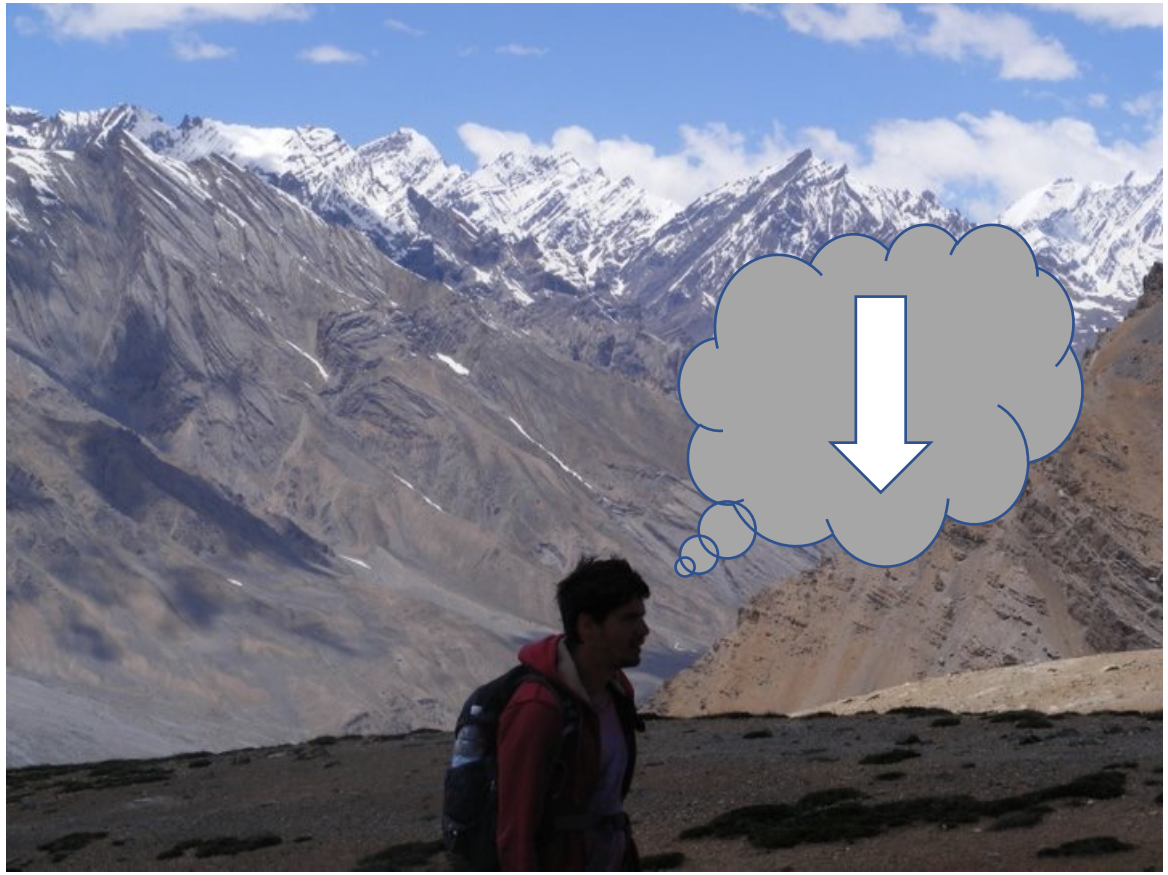
Q: Will normal equations always be practical?

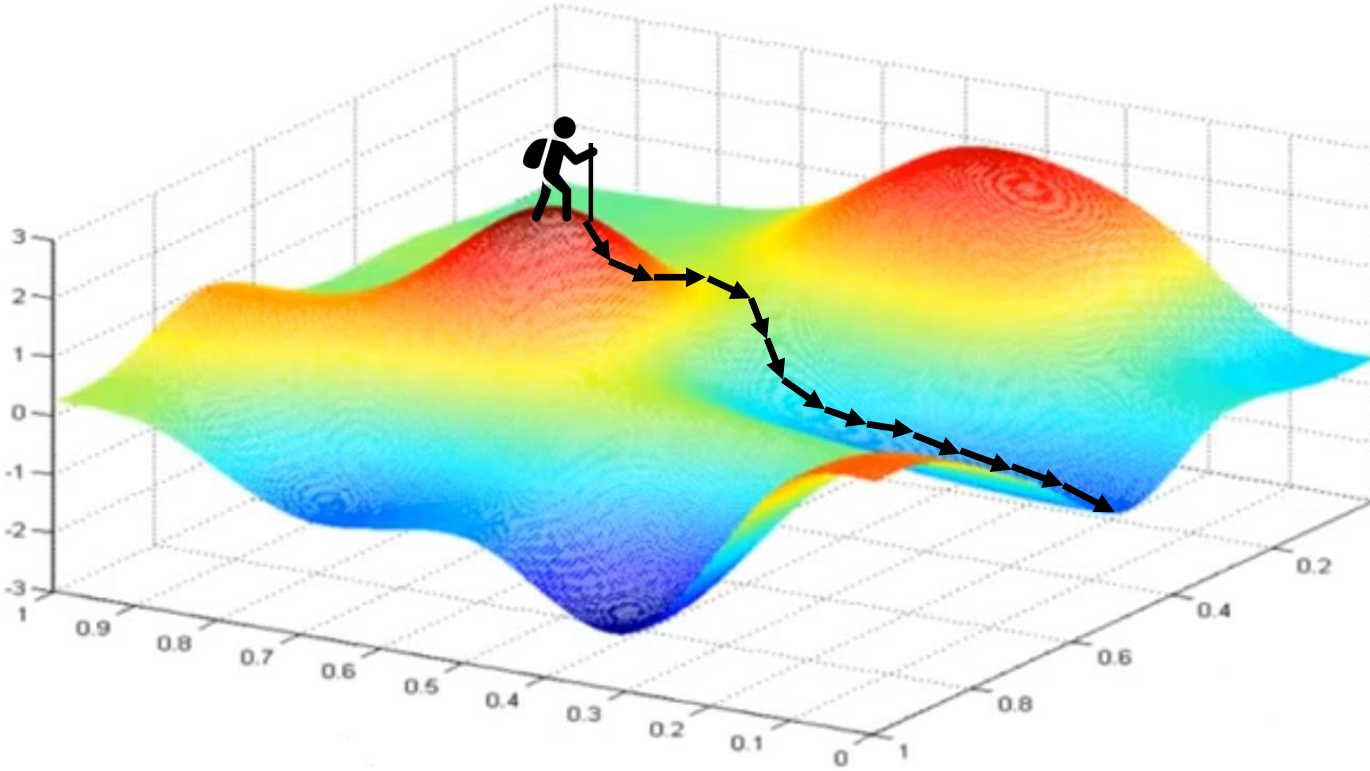
A: No;

1. Inverting $\boldsymbol{X}^T\boldsymbol{X}$ may cost unreasonable memory / time
2. Sometimes not applicable: Regularization? Different loss? Non-linearity?

Gradient descent

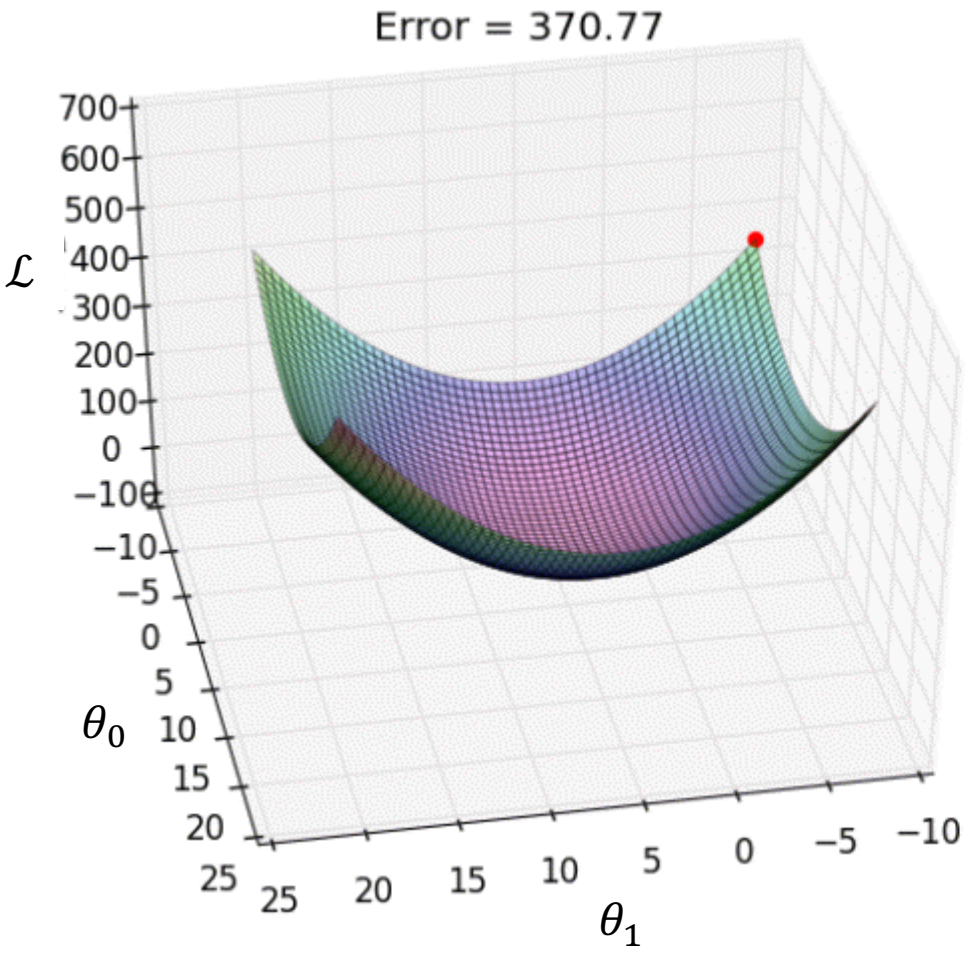
Possible solution: Iteratively reduce loss

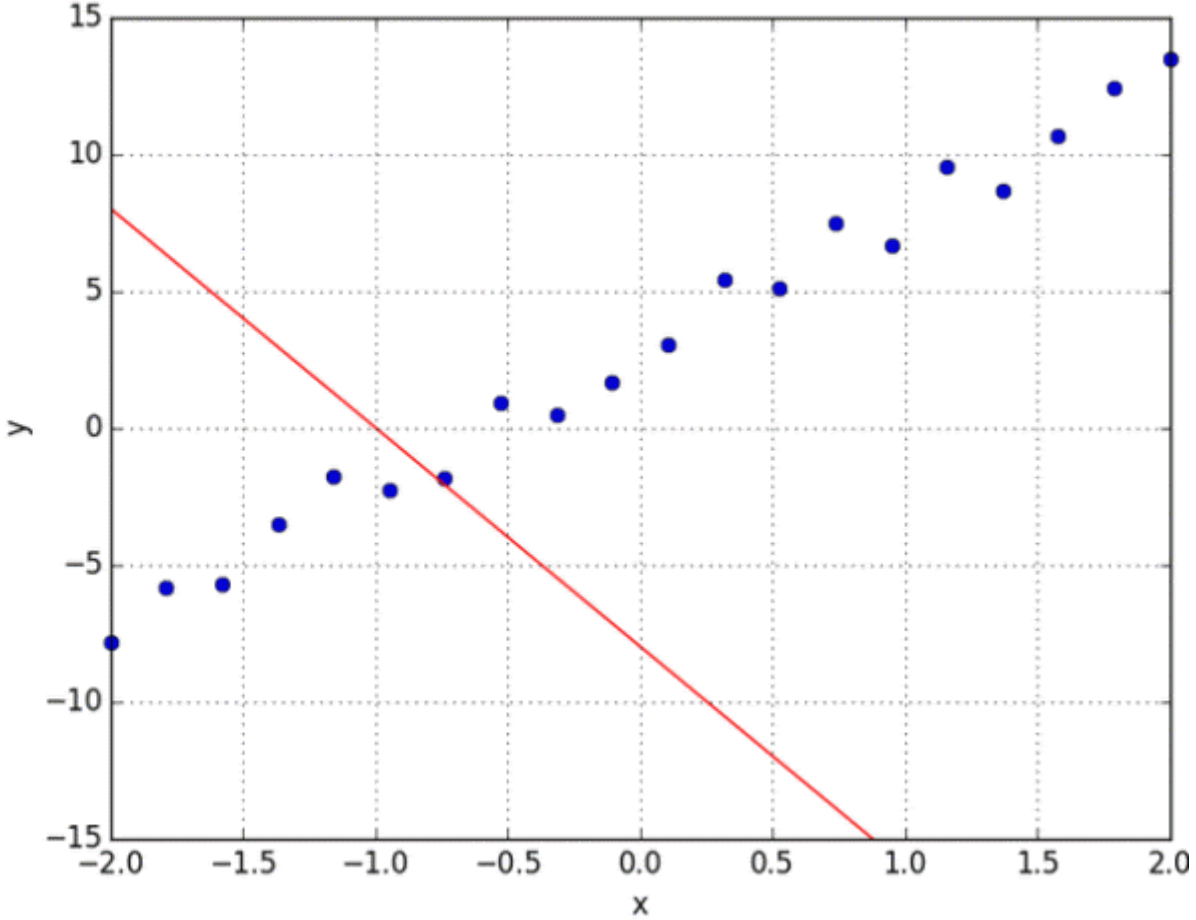




Can we guarantee global min?

Gradient descent



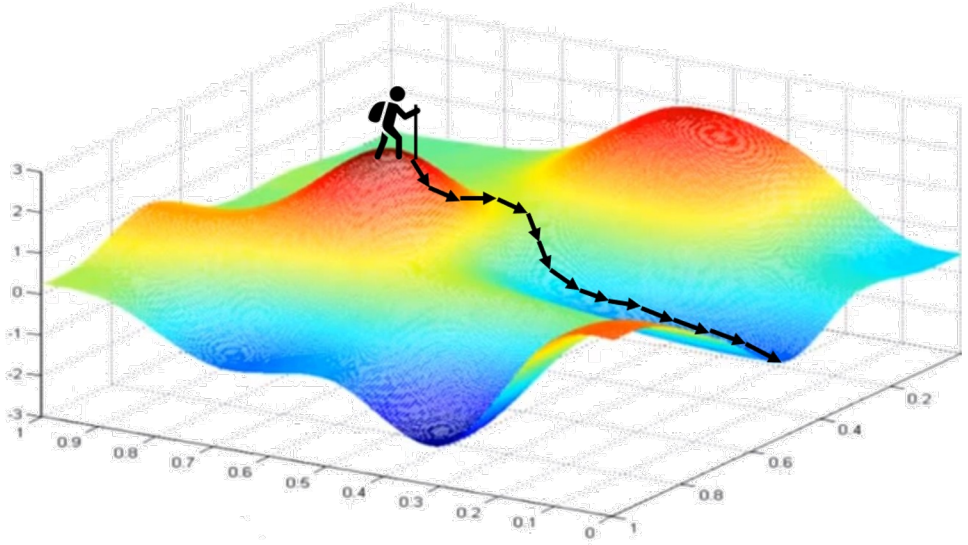


Parameter space: $\mathcal{L}(\boldsymbol{\theta}; S)$

Data space: $h_{\boldsymbol{\theta}}(\boldsymbol{x})$

Gradient descent

$$\boldsymbol{\nabla}\mathcal{L}(\theta_0, \theta_1 \dots \theta_N) = \begin{pmatrix} \frac{\partial \mathcal{L}}{\partial \theta_0} \\ \frac{\partial \mathcal{L}}{\partial \theta_1} \\ \vdots \\ \frac{\partial \mathcal{L}}{\partial \theta_N} \end{pmatrix}$$



Augustin Louis Cauchy

1. Initialize $\theta \sim$Random
2. Repeat until convergence:

$$\{$$

$$\boldsymbol{\theta} := \boldsymbol{\theta} - \alpha \boldsymbol{\nabla}\mathcal{L}(\boldsymbol{\theta}; S)$$

$$\}$$

α: Learning rate

Gradient descent for Linear Regression

$$\mathcal{L} = \frac{1}{2m}\sum_{i=1}^{m}(\boldsymbol{\theta}^T\boldsymbol{x}_i - y_i)^2$$

$$\frac{\partial}{\partial\boldsymbol{\theta}}\mathcal{L} = \frac{1}{m}\sum_{i=1}^{m}(\boldsymbol{\theta}^T\boldsymbol{x}_i - y_i)\boldsymbol{x}_i = \frac{1}{m}\sum_{i=1}^{m}\boldsymbol{x}_i(\boldsymbol{X\theta} - \boldsymbol{y})_{\boldsymbol{i}} = \boldsymbol{X}^T\overbrace{(\boldsymbol{X\theta} - \boldsymbol{y})}^{\boldsymbol{e}}$$

Repeat until convergence:

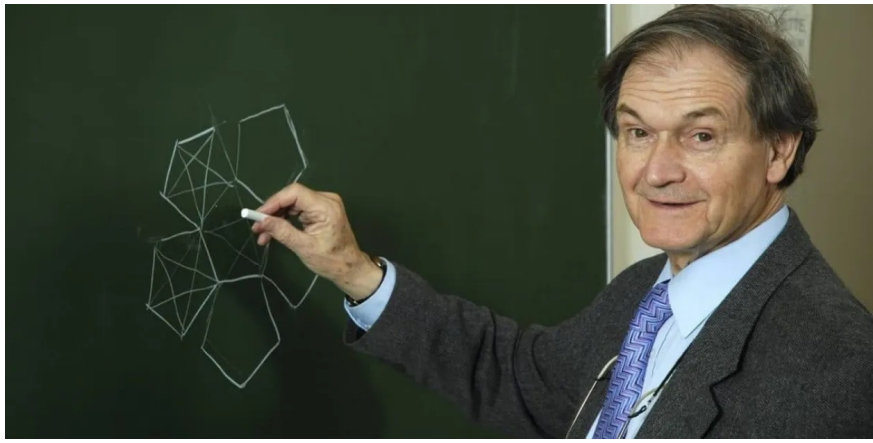
{

$$\boldsymbol{\theta} := \boldsymbol{\theta} - \frac{\alpha}{m}\boldsymbol{X}^T\boldsymbol{e}$$

}

Q: Find the relation between convergence and Normal Equations

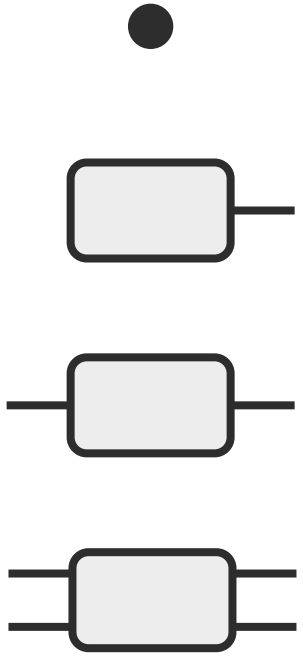
Penrose graphical notation



tensor = box

index = leg

wire = contraction



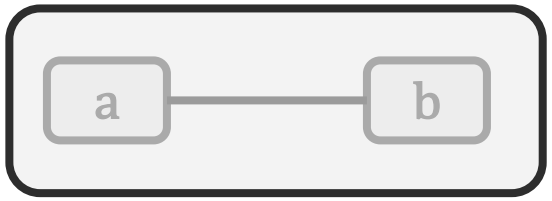
scalar (order 0)

vector (order 1)

matrix (order 2)

tensor (order 4)

inner product



$a_i\, b_i$

→ *scalar (no free legs)*

order 0

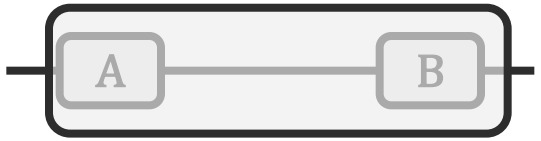
outer product

$a_i\, b_j$

→ *matrix (2 free legs)*

order 2

Penrose graphical notation



mat. multiply

$A_{ij} B_{jk}$

mat. outer product

$A_{ij} B_{kl}$

mat. scalar product (Forbenious)

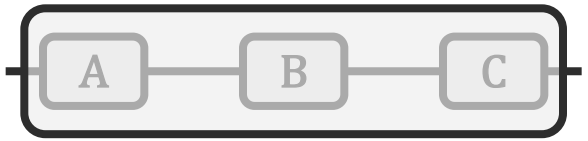
$A_{ij} B_{ij}$

mat. trace

A_{ii}

mat-vec product

$A_{ij} v_j$



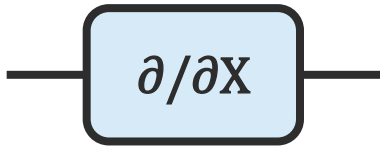
mat. triple chain

$A_{ij} B_{jk} C_{kl}$

<u>Prove (visually)</u>:

Flatten 2 matrices to be vecs and take scalar product = *Multiply the matrices and take trace*

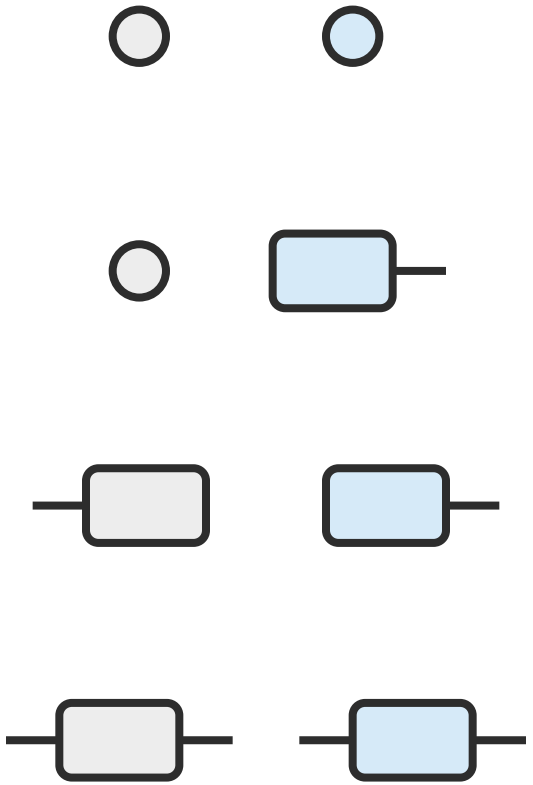
The differentiation operator



Differentiate = outer prod

$$\partial f/\partial X = f \otimes \partial/\partial X$$

$\mathrm{order(J)} = \mathrm{order(f)} + \mathrm{order(X)}$



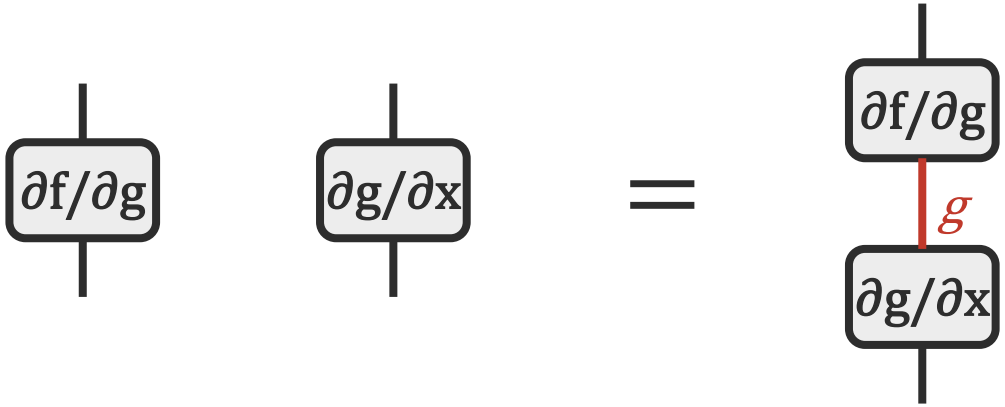
df/dx, scalar

gradient, order 1

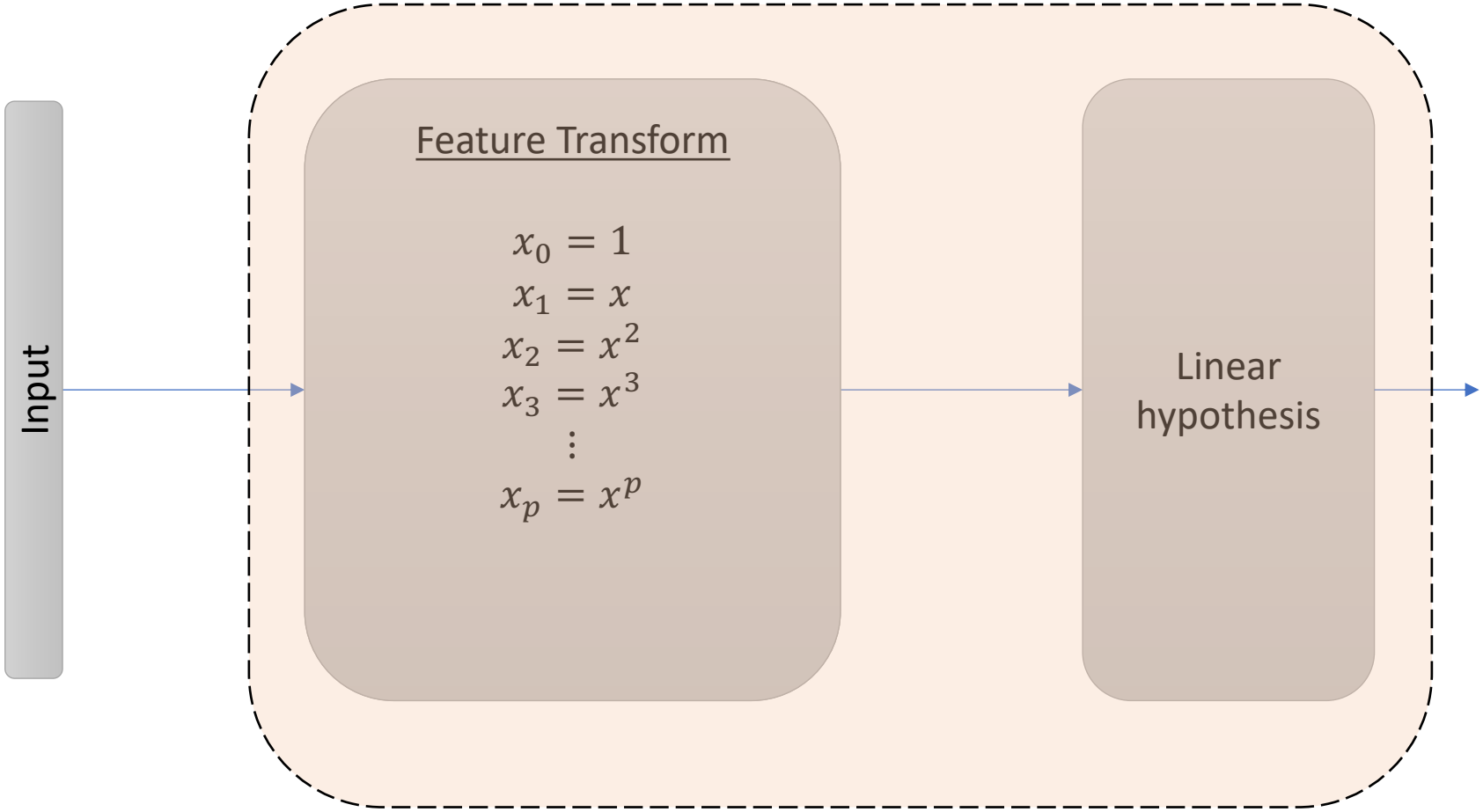
Jacobian matrix, order 2

order-4 Jacobian

Chain rule = inner prod



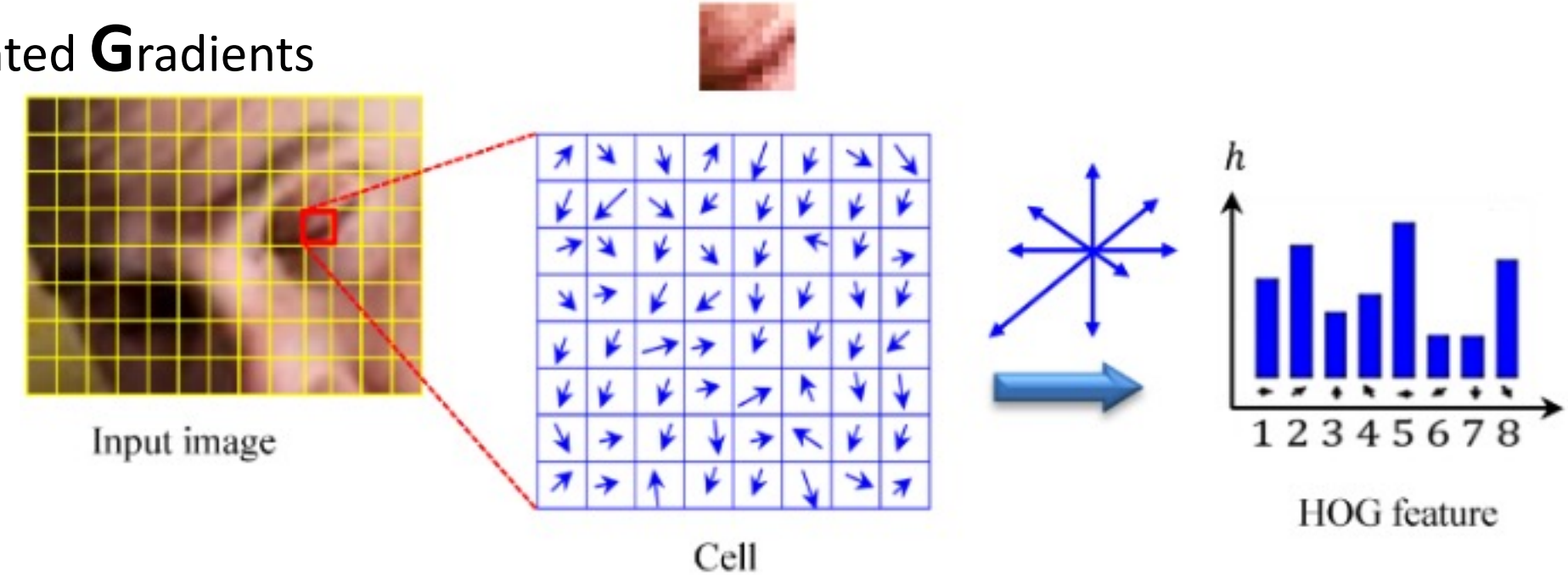
Feature transform



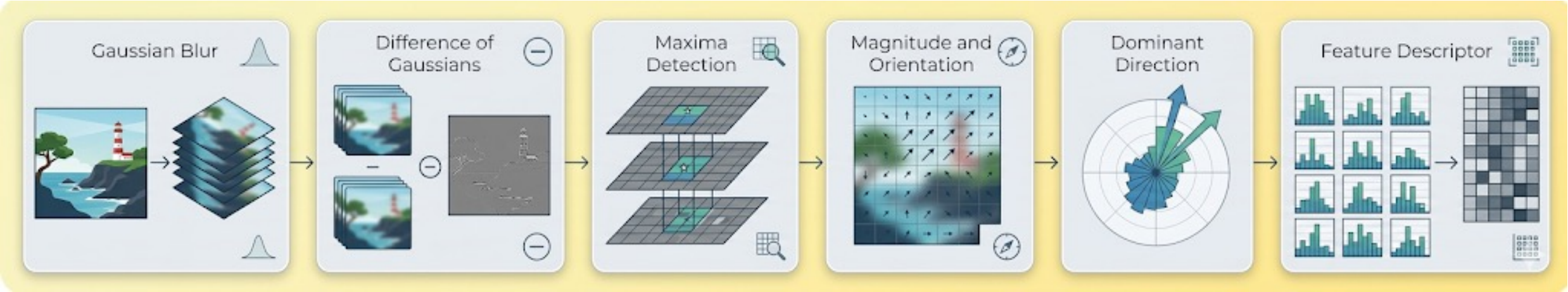
Non-linear hypothesis!
(Polynomial)

Classical image features

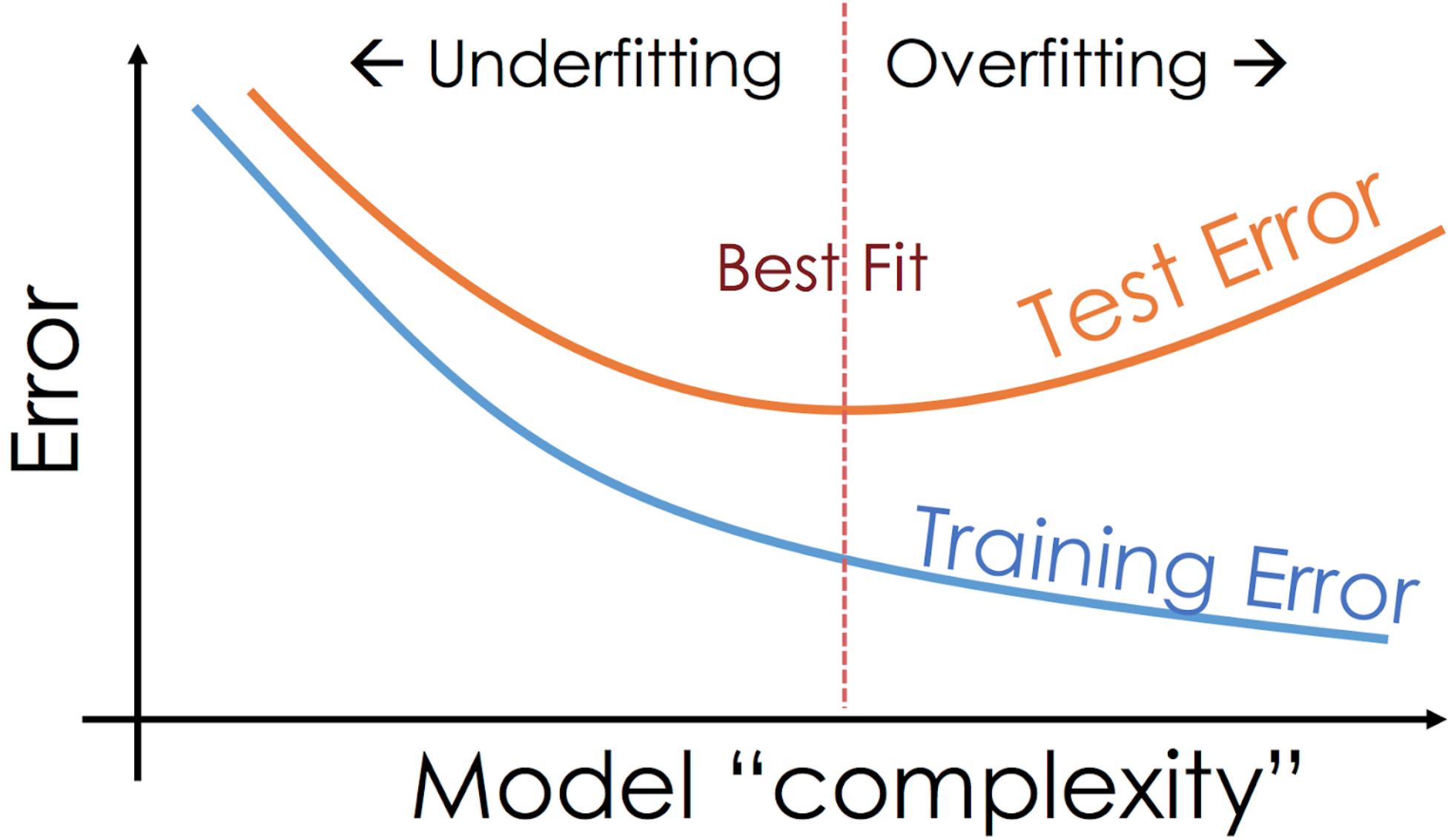
Histogram of **O**riented **G**radients



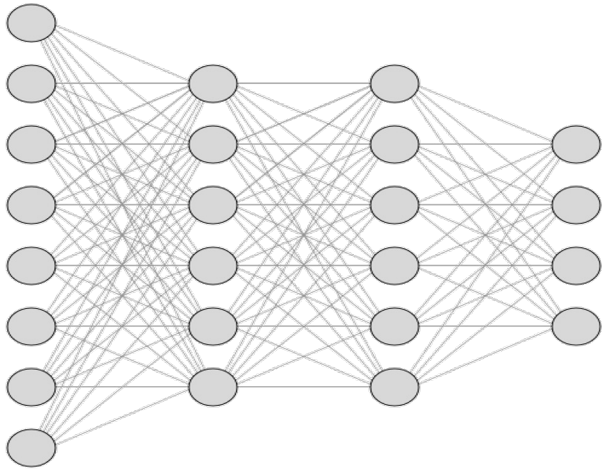
Scale **I**nvariant **F**eature **T**ransform



Generalization



Thank you!

Next week:

Neural Networks